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Prognostic Significance of Serum Complement Activation, Neutrophil Extracellular Traps and Extracellular DNA in Newly Diagnosed Epithelial Ovarian Cancer

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Declaration of interests

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Abstract

Purpose: We observed that the tumor microenvironment (TME) in metastatic epithelial ovarian cancer (EOC) and in other solid tumors can reprogram normal neutrophils to acquire a complement-dependent suppressor phenotype characterized by inhibition of stimulated T cell activation. This study aims to evaluate whether serum markers of neutrophil activation and complement at diagnosis of EOC would be associated with clinical outcomes.

Experimental Design: We conducted a two-center prospective study of patients with newly diagnosed EOC (N = 188). Blood and ascites fluid were collected at diagnosis for biomarker analysis. Patients were evaluated for progression-free survival (PFS) and overall survival (OS).

Results: The median OS was 47 months (95% CI: 34–58) and the median PFS was 12 months (95% CI: 11–15). Pre-treatment serum levels of genomic DNA (gDNA), markers of neutrophil degranulation [myeloperoxidase (MPO)] and neutrophil extracellular traps (NETs) [citruinated histone H3 (CitH3)], and complement activation (C3b/c) were each associated with worse OS on univariate analysis. In multivariate analyses controlling for age, stage, and optimal debulking, serum gDNA, MPO, and CitH3 remained associated with worse OS, while C3b/c levels were not. In an exploratory analysis, the largest magnitude of difference in 2-year OS occurred in patients with low C3b/c and low CitH3 compared to all other patients (87% vs 46% survival, respectively). In ascites fluid, increased factor H, a negative regulator of complement activation, was associated with improved OS on univariate analysis.

Conclusions: These results point to serum gDNA, NETs, and complement activation as potential prognostic biomarkers in patients with newly diagnosed EOC.

INTRODUCTION

Ovarian cancer is frequently metastatic at diagnosis and in the U.S. has a 5-year relative survival of only 50.8% [1]. Epithelial ovarian cancer (EOC) accounts for about 90% of ovarian tumors, the majority being high-grade serous (HGSOC). Tumor-infiltrating T cells are associated with improved outcomes in EOC [2]. However, single-agent immune checkpoint inhibitors (ICI) in recurrent or refractory EOC have response rates of only approximately 10% [3], a finding that points to barriers to T cell expansion and activation in the tumor microenvironment (TME) that limit the effectiveness of immunotherapy. Similar to other tumors, the TME in EOC contains products that obstruct durable T cell-driven anti-tumor immunity, including immunosuppressive myeloid cells [4] and regulatory T cells [5, 6]. We previously observed that neutrophils contribute to inducing a T cell suppressor phenotype when they are exposed to ascites fluid from patients with newly diagnosed EOC and to malignant effusions from other tumors [7–9]. Neutrophil suppressor function was dependent on a number of neutrophil effector functions, including complement signaling [7, 8].

The complement system defends against pathogens, enhances immune responses and promotes inflammation, while excessive complement activation can result in microangiopathy, organ injury, and autoimmunity. Complement activation has a broad effect on tumor cells, immune cells, and vasculature in the TME that can promote or limit tumor growth [10]. For instance, an antitumor effect of complement includes complement-dependent cytotoxicity of tumor cells. However, multiple studies in tumor-bearing mice demonstrate that complement signaling can promote tumor growth and obstruct antitumor immunity [11]. Genetic and pharmacologic inhibition of complement reduced tumor growth through pathways dependent on VEGF and neovascularization in murine EOC [12]. The soluble complement regulator factor H (FH) maintains the balance of complement activation in circulation and on host cells by acting as a cofactor for factor I, promoting the conversion of C3b into proteolytically inactive iC3b. FH also competes with factor B for binding to C3b, thereby preventing the formation of and accelerating the decay of C3 and C5 convertases. Humans have five FH-related (FHR) proteins that are paralogs of FH but lack the domains regulating C3. Therefore, FHRs may act as antagonists of FH through competition for its ligands, potentially promoting local complement activation instead of inhibition. In cancer, the role of FH and FHRs seems to be context-dependent, with both pro-tumor and anti-tumor functions being described. While complement FH (CFH) is highly expressed in various types of cancer and may protect tumor cells from complement attack, the expression of CFHRs is generally low or absent [10].

Complement activation can chemoattract and activate neutrophils, induce neutrophil extracellular traps (NETs) and stimulate thrombosis [13, 14]. NETs are composed of strands of DNA-histone complexes and proteins released by activated neutrophils that ensnare and kill extracellular pathogens. NETs can be injurious and have been implicated in the pathogenesis of autoimmunity, venous thrombosis, and ischemia-reperfusion injury. NET generation accelerated metastasis in several tumor-bearing mouse models, an effect that is mediated by thrombosis and direct signaling to tumor cells [15]. NETs can also enable

tumor cell evasion of T cells and be a barrier to ICI therapy [16, 17], and supported a pre-metastatic omental niche in murine syngeneic EOC [18].

We previously proposed a model in which products of cellular injury (damage-associated molecular patterns [DAMPs]) and pro-inflammatory cytokines, chemokines, and complement in the TME recruit neutrophils and induce a phenotype that can drive tumor progression, including stimulation of NETs, thrombosis, and acquisition of T cell suppressor function [7–9]. In the current study, we tested the hypothesis that complement activation, extracellular DNA (a marker of cell damage) and neutrophil activation and NETs at EOC diagnosis would be associated with worse clinical outcomes.

MATERIALS AND METHODS

Patients

This study was approved by the Institutional Review Boards of Roswell Park Comprehensive Cancer Center (Roswell Park) (Buffalo, NY) and the UPMC Hillman Cancer Center (HCC) (Pittsburgh, PA), and followed federal and state requirements. All participants gave informed consent. Patients with newly diagnosed epithelial ovarian, primary peritoneal, or fallopian tube carcinoma (collectively referred to as EOC) were enrolled from May 2014 through August 2020 and followed through March 2022. We included patients with stage III-IV EOC who had samples collected prior to any treatment. Patients with stage I-II disease, non-epithelial histology, borderline tumors, or low-malignant potential tumors were excluded.

Retrospective review and abstraction of the medical records of Roswell Park patients and de-identified records of HCC patients were performed by two authors at Roswell Park (JR and JJ). Demographics included age, race, smoking history, and body mass index (BMI) at diagnosis. Initial treatment (primary cytoreduction or neoadjuvant chemotherapy [NACT]) and pre-treatment CA 125 were recorded. Pathology reports from primary surgery or diagnostic biopsy provided histology, differentiation, and FIGO stage. For patients who received NACT or never underwent surgery, clinical stage was determined based on clinic notes, initial imaging, and biopsy results. Operative reports were reviewed for the extent of cytoreduction. When not explicitly stated, the description of residual disease in the report was used to determine whether optimal resection was achieved (<1cm of visible disease remaining). The presence of ascites was based on operative report or pre-treatment imaging and was recorded as large or small volume based on the description.

Medical records were retrospectively reviewed for PFS and OS. PFS was defined as the interval between diagnosis and disease progression, recurrence, or death. Date of diagnosis was assigned as the day when the first pathologic evidence of carcinoma was identified either by tissue specimen or ascites fluid. Date of disease recurrence was determined based on the imaging results, surgical findings, or biopsy results that established recurrent disease. If these factors were not available and the clinic note described recurrence based on tumor markers, the date of CA 125 elevation to twice the upper limit of normal was used to determine the date of recurrence [19]. The response to primary treatment was determined

from CT or PET scans obtained after the completion of initial chemotherapy and surgery. OS was the interval from diagnosis to date of death or censored at date of last follow-up.

Sample Collection and Processing

Pre-treatment blood and ascites fluid were collected from patients with newly diagnosed EOC. Additional details on sample collection and processing are contained in the Supplemental Methods.

Quantification of Biomarkers

Quantification of mitochondrial (mtDNA) and genomic (gDNA) DNA in serum and ascites fluid supernatant was performed as previously described [9]. Additional details on DNA extraction and quantification are contained in the Supplemental Methods. Myeloperoxidase (MPO) is located in neutrophil primary granules, is released extracellularly after degranulation, and is a constituent of NETs. Citrullinated histone H3 (CitH3) has been proposed as a biomarker for NET formation, and high plasma CitH3 was associated with an increased risk of cancer-associated thrombosis [20]. Details on measurement of serum and ascites concentrations of MPO and CitH3 are contained in the Supplemental Methods. All complement protein bioanalysis was performed at Sanquin Research (Amsterdam, the Netherlands), using in-house developed antibodies and assays. Complement activation products of C3 (C3b, iC3b and C3c, referred to as C3b/c), C4 (C4b and C4c, referred to as C4b/c), and total levels of FH and FHR proteins were assessed by ELISAs as previously described with minor modifications [21–25], described in the Supplemental Methods.

Statistical Analysis

Biomarker measurements that were right tailed were log transformed. Univariate and multivariate Cox proportional hazard regression analyses were conducted to identify biomarkers that correlate with PFS and OS, as well as to assess whether relationships between established prognostic clinical factors and survival outcomes were behaving as expected in our dataset. Study site (Roswell Park or HCC) was used as a stratum. Multivariate analysis included established prognostic variables for survival: age, FIGO stage (IIC or IV), and debulking status (optimal or suboptimal). Subsequently, we stratified the level of biomarkers as low and high sets using observed median values except C3b/c, where 254nM was selected as the cutoff based on the average of a healthy donor pool. We further evaluated the combination effect of complement proteins, neutrophils degranulation/NETs, and extracellular DNA levels as biomarkers of PFS and OS. Three-group categorization of the combinatory biomarkers was used first to identify the association with OS and PFS, and then the binary stratifications of the combinatory biomarkers (better prognostic subset versus others) were further evaluated. Our primary hypothesis is that complement activation, extracellular DNA and neutrophil activation and NETs at EOC diagnosis would be associated with worse clinical outcomes. Other analyses, including the effect of biomarker combinations, are considered to be exploratory.

Median OS and PFS were calculated for each sub-category of biomarker combinations. When medians were not reached, estimated two-year survival rates were calculated. Kaplan-Meier curves showed survival differences between biomarker groups. Hazard ratios and

p-values from multivariate Cox regression analysis were reported. All statistical analyses were performed using the R 4.2.2 statistical computing language. A nominal significance threshold of 0.05 and 95% confidence interval (CI) estimates were used, unless otherwise specified.

RESULTS

Analysis of Clinical Variables at EOC Diagnosis and Outcome

A total of 188 patients with newly diagnosed EOC were included in the analysis, with descriptive characteristics shown in Table 1.

Patients with optimal debulking had statistically significantly better OS (HR=0.57, 95% CI: 0.37–0.89, $p=0.014$) and PFS (HR=0.65; 95% CI: 0.43–0.97; $p=0.034$), while stage (stage IV versus stage IIIC) showed a trend to worse OS that did not quite reach statistical significance (HR=1.57; 95% CI: 0.97–2.54; $p=0.067$) (Table 2). Ascites was associated with significantly worse PFS (HR=1.81; 95% CI: 1.05–3.13; $p=0.034$). HGSOE had a significantly longer PFS when compared to other types (HR=0.61; 95% CI: 0.41–0.88; $p=0.009$) and a trend towards longer OS (HR=0.67; 95% CI: 0.43–1.03; $p=0.066$).

Serum Levels of gDNA, NETs, and Complement Activation Were Associated with Worse OS in Univariate Analysis

We observed a batch effect for multiple biomarkers in which serum levels were greater in the Roswell Park than HCC cohorts (Supplemental Figure 1). Since the two cohorts were similar regarding clinical variables and outcome, we speculate that this batch effect is related to sample processing, despite implementing the same processing and cryopreservation methods. The two cohorts were adjusted as strata in the univariate and multivariate Cox regression model, which assumes that the baseline hazard is stratum-specific and a common hazard ratio for each covariate is aggregated over strata.

Higher levels of gDNA were associated with significantly shorter OS (HR = 1.19, 95% CI: 1.07–1.31, Cox $p<0.01$) (Table 3). For neutrophil degranulation and NETs biomarkers, statistically shorter OS was found in patients with higher levels of CitH3 or MPO (HR=1.28, 95% CI: 1.09–1.49, $p<0.01$; HR=1.51, 95% CI: 1.16–1.96, $p<0.01$; respectively). MPO was also associated with worse PFS (HR=1.28, 95% CI: 1.02–1.60, $p=0.032$). For complement activation and regulatory proteins, higher levels of C3b/c and FHR-1/2 had significantly worse OS (HR=1.36; 95% CI: 1.04–1.76, $p=0.024$; HR=1.08; 95% CI: 1.01–1.15, $p=0.031$, respectively). Other markers did not have a significant association with OS or PFS on univariate analysis.

Serum Levels of gDNA and NETs are Associated with Worse OS in Multivariate Analyses

In a multivariate Cox regression model that incorporated age, FIGO stages, and debulking status, established prognostic factors [26] (Table 4, Figure 1), serum gDNA remained associated with worse OS (HR = 1.21, 95% CI: 1.08–1.35, Cox $p<0.01$) (Figure 1A). Patients with higher levels of CitH3 had worse OS (HR=1.24; 95% CI: 1.05–1.47, Cox $p=0.01$) and PFS (HR=1.13, 95% CI: 1.01–1.26, Cox $p=0.04$) (Figure 1B and 1C). Higher

levels of MPO remained associated with worse OS (HR=1.57, 95% CI: 1.17–2.09, Cox $p<0.01$) and PFS (HR=1.34, 95% CI: 1.05–1.7, Cox $p=0.017$) (Figure 1D and 1E). After adjusting for confounding variables, C3b/c and FHR-1/2 were not associated with OS.

Pairwise Associations Between Serum C3b/c, NET Formation, and Genomic DNA are Associated with OS

We next evaluated the interaction of complement activation, NETs, and gDNA biomarkers at diagnosis with patient outcomes in a subset of 106 patients with one or more pairs of markers. The 2-year PFS and OS were 24% (95% CI: 16%–35%) and 66% (95% CI: 57%–76%), respectively. Stratifying patients into high and low groups, we considered the combined levels of C3b/c and complement-related biomarkers (FH, FHR-1/1 and FHR-1/2), markers of neutrophil degranulation and NETs (MPO, CitH3), and cellular damage (gDNA) separately. The median PFS and OS were 12 months (95% CI: 10–15) and 47 months (95% CI: 37–59), respectively (Supplemental Table 1). We observed strong favorable OS trends for patients with low levels of C3b/c and low levels of CitH3; low levels of C3b/c, high levels of FH, and low levels of CitH3. The combination of low levels of C3b/c and low levels of MPO was correlated with better PFS.

Lower Serum Levels of C3b/c Combined with High Levels of FH and Lower levels of CitH3, MPO, and gDNA are Associated with Improved Survival

We next built multivariate Cox proportional hazard models for the dichotomized combination of complement activation (C3b/c) with other biomarkers, adjusted for age, FIGO stage, and debulking status. Median PFS for patients with lower levels of C3b/c (<254 nM) and higher levels of FH (> median) was 21 months, versus 11 months for the rest of the population (HR= 0.46, 95% CI:0.21–1, Cox $p= 0.049$) (Figure 2a). The 2-year PFS for patients with low C3b/c and high FH was 44% (95% CI 25%–78%) versus 13% (95% CI 6%–30%) for all other patients (Supplemental Table 2). Longer OS was observed in patients with lower levels of C3b/c and lower levels of CitH3 (< median) than in other patients, with 2-year OS of 87% (95% CI 75%–1.00%) and 46% (95% CI 32%–64%), respectively (HR= 0.27, 95% CI:0.10–0.68, $p<0.01$) (Supplemental Table 2, Figure 2b). Lower levels of C3b/c and lower levels of MPO (< median) were associated with better PFS, with 2-year PFS 35% versus 8% (HR=0.45, 95% CI: 0.22–0.93) (Supplemental Table 2, Figure 2c). Patients with lower levels of C3b/c and lower levels of gDNA (< median) had better OS than other patients, with 2-year OS 81% versus 51% (HR= 0.31, 95% CI:0.11–0.83, $p= 0.02$) (Supplemental Table 2, Figure 2d).

Ascites Fluid FH Levels are Associated with Improved PFS and OS

We next applied the same biomarker analysis to ascites fluid in 134 patients. In paired samples, serum levels exceeded ascites fluid levels of all biomarkers except C3b/c and mtDNA (Supplemental Table 3). HGSOc histology and optimal debulking were associated with improved PFS, and optimal debulking was associated with improved OS. In the univariate analysis, ascites fluid FH levels were associated with improved PFS (HR= 0.67, 95% CI: 0.46–0.97, Cox $p= 0.032$) and OS (HR= 0.53, 95% CI:0.33–0.85, Cox $p<0.01$), and CitH3 levels were associated with improved OS (HR= 0.86, 95% CI:0.75–0.98, $p= 0.029$) and a trend toward improved PFS. Ascites fluid FHR-5 levels were associated with

improved OS (HR= 0.73, 95% CI:0.57–0.94, Cox p= 0.013) but not PFS, and ascites fluid MPO levels were associated with improved PFS (HR= 0.81, 95% CI:0.67–0.99, Cox p= 0.041) (Supplemental Table 5). In a multivariate analysis that incorporated relevant clinical variables, ascites fluid MPO levels remained significantly associated with improved PFS (HR= 0.76, 95% CI:0.62–0.94, Cox p= 0.011), and ascites fluid FH was associated with a trend to improved PFS (Supplemental Table 6).

Serum CitH3, gDNA, and FHR-1/2 are Associated with Worse OS in Patients with HGSOC

In a separate analysis of patients with HGSOC (Supplemental Tables 7–9), optimal debulking was associated with improved OS in univariate analysis. Serum CitH3, gDNA, and FHR-1/2 were each associated with worse OS in univariate and multivariate analysis.

DISCUSSION

Serum levels of circulating gDNA, markers of neutrophil degranulation and NETs (MPO and CitH3), and complement activation (C3b/c) at EOC diagnosis were each associated with worse OS in univariate analyses. In the multivariate analyses where we controlled for historically established prognostic variables (age, stage, optimal debulking), serum gDNA, MPO, and CitH3 remained associated with worse OS, while C3b/c levels were not independently associated with worse OS. To further refine the prognostic value of these markers, we conducted an exploratory analysis of the association between serum C3b/c paired with gDNA, MPO, or CitH3 and outcome. The largest effect appeared at 2 years, where the survival rate was 87% for patients with low C3b/c and low CitH3 versus 46% in other patients (Supplemental Table 2). Patients with lower levels of C3b/c and lower levels of gDNA also had better OS than other patients (Supplemental Table 2), and the combination of lower levels of C3b/c and lower levels of MPO was associated with better PFS (Supplemental Table 2).

In the exploratory pairwise analysis, we observed improved OS in patients with low C3b/c and high FH levels. Consistent with our results of C3b/c in serum, others have observed circulating C4d, a complement activation product, to be associated with poor prognosis in patients with solid tumors [27–29]. In contrast, we did not observe an association between the complement activation fragment C4b/c in ascites of EOC patients and survival. Since FH is a negative regulator of alternative pathway complement activation, these results support the notion of the association of complement activation and worse prognosis. We observed a weak but statistically significant association between FHR-1/2 and worse OS (Table 3) that was not significant in the multivariate analysis (Table 4). This finding aligns with the hypothesis that the FHR proteins compete with FH for ligand binding and subsequently promote complement activation [30]. Because FHR-1 and -2 form both hetero- and homodimers, it is interesting that only FHR-1/2 was associated with clinical outcome. So far, only FHR-1 homodimers and FHR-1/2 heterodimers can be directly determined, while assays for the direct quantification of FHR-2/2 homodimers are lacking. With only FHR-1/2 being clinically associated, results suggest FHR-2 could be the main driver in this association. The dimer distribution may play a role in the (de)regulation of the complement system.

Patients with HGSOC had longer PFS and a trend towards longer OS when compared to other types. HGSOC usually has a worse prognosis than other EOC histologies, and therefore our observation may be a chance finding. In addition, our patient cohort was limited to stage IIIC and IV EOC, and about half had a large volume of ascites fluid; these factors are expected to be associated with decreased survival across histologic subtypes. When our analysis was limited to patients with HGSOC, we observed similar findings compared to the analysis of all patients, with serum CitH3, gDNA, and FHR-1/2 each associated with worse OS.

Part of the rationale for our study is based on our findings that neutrophils acquire T cell suppressor function in the TME, an effect that was dependent on a number of neutrophil effector functions, including complement signaling [7, 8]. There are numerous additional pathways by which neutrophils can accelerate tumor progression, including stimulation of thrombosis and neovascularization, extracellular matrix remodeling, and direct signaling to tumor cells [31]. NET generation can also stimulate tumor progression through thrombosis, tumor cell signaling, and impairing T cell immune attack of tumor cells. Though we posit that in most contexts, activated complement and neutrophils drive tumor progression, both may also have beneficial effects that include complement-dependent cytotoxicity of tumor cells, neutrophil-driven tumor cell killing, and tumor-associated neutrophils stimulating T cell responses [32]. We therefore underscore the importance of unraveling specific cues in the TME and anti-tumor therapies that influence whether complement activation and neutrophils are foes or friends.

Consistent with our results in EOC, Revel et al. [33] observed that elevated plasma levels of C4d (a complement activation product) and Bb (required for alternative pathway activation) at the time of surgery correlated with worse prognosis in patients with clear cell renal cell carcinoma. They proposed “Humoral Complementomics” consisting of complement proteins, activation fragments, and autoantibodies, as a tool to evaluate systemic complement activation that could be leveraged as prognostic biomarkers [33]. Tumor cells can produce complement proteins and complement regulatory proteins with autocrine and paracrine signaling. Cho et al. [34] observed that tumor cell-derived complement can induce autocrine signaling that drives tumor cell proliferation, and that higher tumor C3 or C5aR mRNA levels were associated with decreased overall survival in patients with ovarian or lung cancer. The location of complement activation, both systemic and in the TME, and the source of complement products can have diverse effects on tumor cell signaling and modulation of the immune landscape. In addition, complement signaling may have important interactions with other tumor-promoting genes that drive chemotherapy resistance and poor outcomes in EOC [35].

We used circulating cell-free DNA as a marker of cellular damage without regard to the source. Several studies have shown that the DNA integrity index measured as the ratio of long to short DNA fragments (an estimate of necrotic versus apoptotic cell death, respectively) is higher in patients with various cancers and is potentially applicable for cancer screening and as a prognostic biomarker [36–38]. Extracellular DNA is a DAMP released during injury that stimulates inflammation and injury [39]. In addition, NETs are also a potential source of circulating extracellular DNA [40].

Zhang et al. [41] developed a “NETs score” calculated based on 19 NETs-related genes and observed that the NETs score was associated with poor prognosis in patients with most solid tumors. Circulating NET markers are elevated in patients with advanced solid tumors and have been associated with increased mortality [42–44]. Kim et al. [45] observed that patients with HGSOC had higher levels of the three NET markers (histone-DNA complex, cell-free DNA, and neutrophil elastase) and prekallikrein than healthy controls, and neutrophil elastase was independently associated with worse OS. In contrast, Dobilas et al. [46] observed that plasma levels of CitH3-DNA and dsDNA were not elevated in women with borderline or malignant ovarian tumors compared with those with benign adnexal masses and did not correlate with OS among patients with malignant tumors.

The prognostic relationships of biomarkers in ascites fluid were different than in serum. C3b/c and mtDNA levels were greater in ascites fluid than serum, while serum levels of other biomarkers exceeded ascites levels. Ascites fluid levels reflect diffusion of products from plasma and local production and degradation of these products in the TME. Ascites fluid FH levels were associated with improved PFS and OS on univariate analysis. Although this observation supports a beneficial role of FH in limiting complement activation in the TME, it is also possible that FH and other complement regulatory proteins could protect tumor cells from complement attack [47]. In contrast to our prior observations from a separate cohort of patients with EOC showing that ascites mtDNA levels were associated with worse outcomes [9], we didn’t observe a prognostic effect of either ascites mtDNA or gDNA on outcome in the current study. The discrepant findings might relate to differences in processing and that the prior study was a single-center retrospective analysis involving a smaller number of patients.

Similar to other solid tumors, paraneoplastic thrombocytosis is associated with worse outcomes in metastatic EOC [48, 49]. Circulating and extravasated platelets can drive tumor progression in a number of ways, including supporting a pre-metastatic niche for hematogenous seeding [50] and promotion of angiogenesis within tumors [51]. Additional research is warranted to delineate the prognostic interactions between thrombocytosis and the proportion of activated platelets in circulation with markers of neutrophil and complement activation in cancer prognosis.

Our findings point to the value of serum markers of cellular damage, NETs, and complement as potential prognostic biomarkers in patients with newly diagnosed EOC. Strengths of our study include the prospective specimen collection, multi-center enrollment, large sample size, and duration of follow-up. Our study was limited to serum and cell-free ascites fluid and didn’t include resected tumors. Though promising as a prognostic tool, our analyses of the association of combination biomarkers and outcomes weren’t pre-specified and should be considered exploratory. The multivariate analysis included age, FIGO stage, and debulking status as confounding variables, and we note the limitation that other confounders related to the tumor (e.g. BRCA status and tumor immune landscape) and host factors (e.g., diet and comorbidities) could have affected our findings. We did not include multiple testing correction, given the small number of novel biomarkers tested and our pre-specified hypothesis that complement activation, NET markers, and extracellular DNA at EOC diagnosis would be associated with worse clinical outcomes. Our study was limited to a

single time point for specimen collection prior to initiation of therapy; the change in these markers in response to therapy could be of prognostic value to guide treatment and warrants further evaluation. In addition, the performance of these biomarkers require validation in an independent cohort of patients with advanced EOC and be tested in patients with other metastatic tumors.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Translational Relevance

Similar to other solid tumors, the tumor microenvironment (TME) in metastatic epithelial ovarian cancer (EOC) is characterized by injury and inflammation. We observed that the TME can reprogram normal neutrophils to acquire a complement-dependent suppressor phenotype characterized by inhibition of stimulated T cell activation. Building on these results, the current study aimed to test the extent to which markers of neutrophil and complement activation would correlate with clinical outcomes in EOC. In a two-center prospective study of patients with newly diagnosed EOC, serum markers of complement activation, neutrophil extracellular traps, and extracellular genomic DNA, a marker of cellular damage, were associated with worse overall survival. Ascites fluid factor H, a negative regulator of complement activation, was associated with improved survival. These results provide indirect support for pathologically activated neutrophils and complement impairing responses to standard treatment for EOC and establish the rationale for further testing these markers as prognostic biomarkers.

- Immune checkpoint inhibitors have been shown to be largely ineffective as a treatment for epithelial ovarian cancer (EOC).
- Neutrophils, NETs, and complement signaling have previously been implicated in inducing T cell suppression.
- In this study, complement activation, NETs, extracellular gDNA (cellular damage marker), associated with worse EOC survival.
- Ascites fluid factor H, a negative regulator of complement activation, was associated with improved survival.

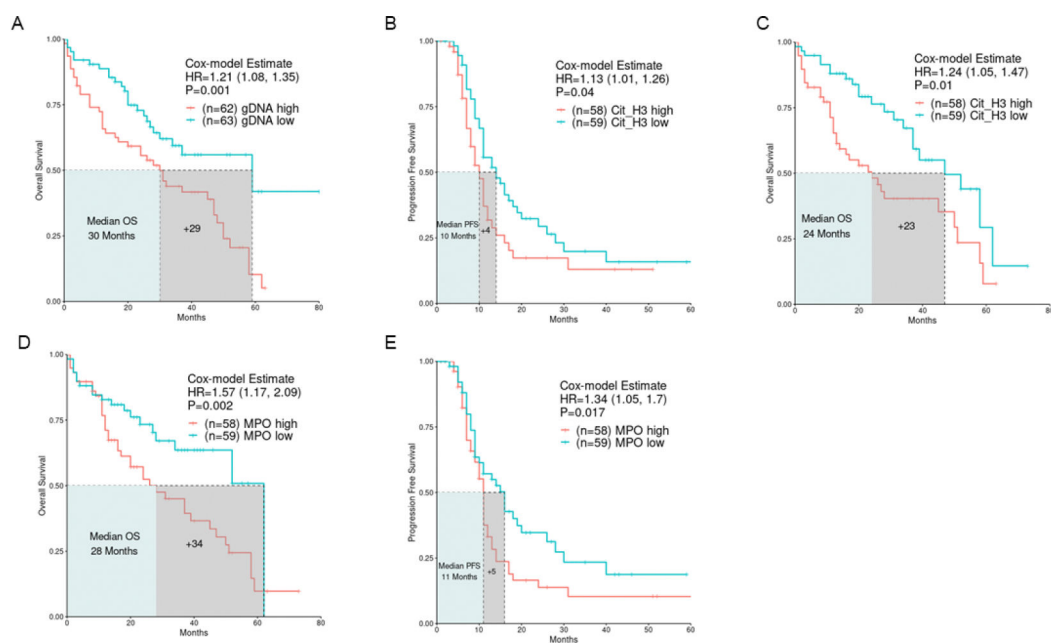


Figure 1. Pre-treatment serum levels of gDNA and NETs are Associated with Worse overall survival in multivariate analyses.

Kaplan-Meier plots for serum levels of (A) gDNA, (B and C) CitH3, and (D and E) MPO in association with OS or PFS. Cox proportional hazard model-based estimates (hazard ratio with 95% confidence interval, p-value) are reported.

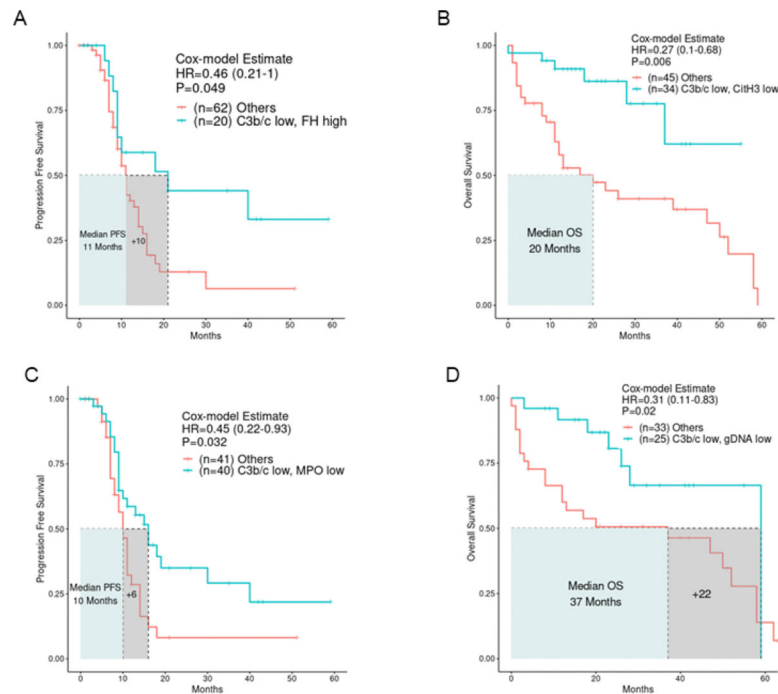


Figure 2. Pre-treatment lower serum levels of C3b/c combined with higher levels of FH and lower levels of CitH3, MPO, and gDNA are associated with improved overall survival. Kaplan-Meier plots for serum levels of C3b/c combined with (A) FH, (B) CitH3, (C) MPO, and (D) gDNA separately. Cox proportional hazard model-based estimates (hazard ratio with 95% confidence interval, p-value) are reported.

Table 1.

Demographic and clinical characteristics of 188 patients with newly diagnosed advanced epithelial ovarian cancer recruited at Roswell Park Comprehensive Cancer Center and UPMC Hillman Cancer Center, May 2014 through August 2020.

Characteristics	
	Mean (SD)
Age at diagnosis (years)	65.6 (12.1)
	Median (IQR)
Serum CA-125	764 (1917)
	Median (95% CI)
Overall survival (OS) (mo.)	47 (34–58)
Progression-free survival (PFS) (mo.)	12 (11–15)
	N (%)
Study Site	
Roswell Park Comprehensive Cancer Center	84 (44.7)
UPMC Hillman Cancer Center	104 (55.3)
Body mass index (BMI)	
Underweight (<18.5 kg/m ²)	5 (2.7)
Normal weight (18.5<25.0 kg/m ²)	50 (26.6)
Overweight (25.0<30.0 kg/m ²)	65 (34.6)
Obese (≥30.0 kg/m ²)	66 (35.1)
Race	
White	169 (90.0)
Black	10 (5.3)
Other or unknown races	9 (4.8)
Smoking status	
Never	112 (59.6)
Former	46 (24.5)
Current	25 (13.3)
Ascites	
Present	154 (81.9)
Not present	29 (15.4)

Ascites volume	
Small or moderate volume	56 (29.8)
Large volume	89 (47.3)
Unknown volume	9 (4.8)
Primary site	
Ovary	111 (59.0)
Fallopian tube	26 (13.8)
Peritoneum	15 (8.0)
Unknown	36 (19.2)
FIGO Stage	
IIIC	143 (76.1)
IV (NOS)	28 (14.9)
IVA	5 (2.7)
IVB	12 (6.4)
Grade	
1: Well differentiated	8 (4.3)
2: Moderately differentiated	2 (1.1)
3: Poorly differentiated	159 (84.6)
Unknown	19 (10.1)
Histotype	
Serous	143 (76.1)
Mucinous	2 (1.1)
Endometrioid	5 (2.7)
Clear cell	10 (5.3)
Mixed epithelial	9 (4.8)
Other epithelial / unknown, but known to be epithelial	19 (10.1)
Epithelial ovarian carcinoma	
HGSOC	127 (67.6)
LGSOC	6 (3.2)
Other/unknown	55 (29.3)
Type of chemotherapy	
Neoadjuvant	80 (42.6)
Adjuvant only	96 (51.1)
None	9 (4.8)
Unknown	3 (1.6)
Optimal debulking	
Yes	126 (67.0)
No	55 (29.3)
Response to primary therapy	

Complete response	87 (46.3)
< Complete response	76 (40.4)
Still receiving primary chemotherapy at last follow-up	3 (1.6)
Did not complete primary chemotherapy	15 (8.0)
No primary treatment	7 (3.7)

[†]Total categorical values may not sum to 100% due to missing data and rounding.

SD = standard deviation; IQR = interquartile range; CI = confidence interval; N = number; HGSOC = high-grade serous ovarian cancer; LGSOC = low-grade serous ovarian cancer.

Table 2.

Association of clinical variables with overall survival (OS) and progression-free survival (PFS) by univariate Cox regression model.

Clinical Variables	N	PFS			OS		
		N(evt)	HR	P	N(evt)	HR	P
Age (years)	188	123	1.00(0.99, 1.02)	0.597	88	1.01(0.99, 1.03)	0.329
BMI (kg/m ²)	186	121	1.02(1.00, 1.05)	0.101	86	1.00(0.97, 1.03)	0.964
Serum CA-125 (U/mL)	171	114	1.00(1.00, 1.00)	0.337	83	1.00(1.00, 1.00)	0.211
Study Site (Roswell Park vs HCC)	188	123	0.93(0.65, 1.33)	0.672	88	0.86(0.55, 1.36)	0.522
FIGO Stage (IV vs IIIC)	188	123	0.80(0.50, 1.28)	0.344	88	1.57(0.97, 2.54)	0.067
Optimal debulking (yes vs no)	181	120	0.65(0.43, 0.97)	0.034 *	84	0.57(0.37, 0.89)	0.014 *
Ascites (present vs not)	183	121	1.81(1.05, 3.13)	0.034 *	86	1.42(0.72, 2.77)	0.310
EOC histology (HGSOC vs other)	188	123	0.61(0.41, 0.88)	0.009 *	88	0.67(0.43, 1.03)	0.066
Neoadjuvant chemotherapy (yes vs no)	187	122	1.26(0.86, 1.84)	0.230	87	1.25(0.78, 2.02)	0.355
Smoking (former vs never)	183	119	0.98(0.63, 1.51)	0.925	85	0.97(0.58, 1.63)	0.917
Smoking (current vs never)	183	119	1.36(0.81, 2.29)	0.251	85	1.63(0.87, 3.04)	0.127

N: total number of patients with available clinical data. N(evt): total number of events (progression or death) within N.

* P<0.05.

Age, BMI, and serum CA-125 are continuous variables.

Table 3.

Association of serum DNA, complement activation, and NET markers with PFS and OS by univariate Cox regression model.

Biomarkers	N	PFS			OS		
		N(evt)	HR	P	N(evt)	HR	P
CitH3	117	75	1.10 (0.98, 1.23)	0.090	57	1.28 (1.09, 1.49)	0.002 *
gDNA	125	84	1.04 (0.95, 1.14)	0.400	68	1.19 (1.07, 1.31)	0.001 *
MPO	117	75	1.28 (1.02, 1.60)	0.032 *	57	1.51 (1.16, 1.96)	0.002 *
mtDNA	125	84	1.00 (0.93, 1.08)	0.909	68	1.04 (0.95, 1.14)	0.419
C3b/c	107	67	1.04 (0.84, 1.28)	0.737	44	1.36 (1.04, 1.76)	0.024 *
FH	107	67	1.00 (0.99, 1.00)	0.269	44	1.00 (0.99, 1.00)	0.743
FHR-1/1	107	67	1.03 (1.00, 1.06)	0.093	44	1.02 (0.98, 1.06)	0.343
FHR-1/2	105	66	1.04 (0.98, 1.10)	0.202	44	1.08 (1.01, 1.15)	0.031 *
FHR-3	107	67	1.05 (0.89, 1.23)	0.545	44	1.04 (0.85, 1.29)	0.681
FHR-4A	107	67	0.91 (0.69, 1.20)	0.507	44	1.10 (0.81, 1.48)	0.548
FHR-5	107	67	1.04 (0.74, 1.45)	0.825	44	1.12 (0.74, 1.70)	0.596

N: total number of patients available with valid biomarker values. N(evt): total number of events (progression or death) within N.

* P<0.05.

Table 4.

Association of serum DNA, complement activation, and NET markers with PFS and OS by multivariate Cox regression model.

Biomarker	N	PFS			OS		
		N(evt)	HR	P	N(evt)	HR	P
CitH3	111	73	1.13 (1.01, 1.26)	0.040 [*]	54	1.24 (1.05, 1.47)	0.01 [*]
gDNA	120	82	1.05 (0.96, 1.15)	0.328	65	1.21 (1.08, 1.35)	0.001 [*]
MPO	111	73	1.34 (1.05, 1.70)	0.017 [*]	54	1.57 (1.17, 2.09)	0.002 [*]
mtDNA	120	82	1.01 (0.94, 1.10)	0.759	65	1.04 (0.94, 1.15)	0.408
C3b/c	101	65	1.03 (0.83, 1.29)	0.776	41	1.24 (0.93, 1.66)	0.142
FH	101	65	1.00 (0.99, 1.00)	0.120	41	1.00 (1.00, 1.01)	0.839
FHR-1/1	101	65	1.03 (1.00, 1.07)	0.087	41	1.01 (0.96, 1.05)	0.813
FHR-1/2	99	64	1.03 (0.97, 1.10)	0.319	41	1.07 (0.99, 1.15)	0.090
FHR-3	101	65	1.06 (0.89, 1.27)	0.500	41	1.02 (0.82, 1.26)	0.869
FHR-4A	101	65	0.89 (0.67, 1.18)	0.424	41	1.10 (0.80, 1.52)	0.551
FHR-5	101	65	1.01 (0.72, 1.43)	0.947	41	1.12 (0.74, 1.71)	0.588

N: total number of patients available with valid biomarker values. N(evt): total number of events (progression or death) within N.

^{*} P<0.05.